

Body Performance Analytics

Intelligent Classification System Using Machine Learning

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Project Overview

Objective: Predict physical performance levels from body measurements and fitness tests

Dataset: 13,394 individuals with 12 variables (age, gender, body metrics, fitness scores)

Tasks: Classification (A/B/C/D) and Regression (broad jump prediction)

Models: KNN, Decision Tree, SVM, Linear Regression, Neural Network

Validation: Multiple train-test splits (80:20, 70:30, 50:50) + 5-fold cross-validation

Dataset Characteristics

Key Statistics

13,394 total observations

12 variables measured

0 missing values (100% complete)

Age range: 21-64 years

Gender: 68% Male, 32% Female

4 performance classes (A, B, C, D)

Variables Measured

Age, Gender, Height, Weight

Body Fat Percentage

Blood Pressure (Systolic/Diastolic)

Grip Force

Sit-and-Reach Flexibility

Sit-ups Count

Broad Jump Distance

Exploratory Data Analysis Insights

Data Quality: Excellent

No missing values detected

Minimal duplicates removed

All values within physiologically plausible ranges

Strong data integrity confirmed

Key Correlations

Sit-ups $\leftrightarrow$ Broad Jump: $r = 0.71$ (strong positive)

Age $\leftrightarrow$ Grip Force: $r = -0.45$ (moderate negative)

Height $\leftrightarrow$ Weight: $r = 0.62$ (strong positive)

Body Fat $\leftrightarrow$ Performance: Inverse relationship

Classification Models Performance

Model	Accuracy	Precision	Recall	F1-Score
KNN (k=7)	0.89	0.88	0.89	0.89
Decision Tree (depth=10)	0.87	0.86	0.87	0.87
SVM (RBF kernel)	0.91	0.90	0.91	0.91



Best Performer: SVM with RBF kernel achieved 91% accuracy across all metrics

Regression Models Performance

Predicting Broad Jump Distance (cm)

Model	MSE	RMSE	R ² Score
Linear Regression	152.4	12.3	0.82
Neural Network (100,50)	138.2	11.8	0.85



Best Performer: Neural Network

Explains 85% of variance in broad jump performance with average error of only 11.8 cm

Cross-Validation Experiments

Multiple Split Ratios Tested

80:20 Split - Standard evaluation (91% accuracy for SVM)

70:30 Split - More test data (90% accuracy, <1% variation)

50:50 Split - Equal division (89% accuracy, consistent performance)

5-Fold CV - Comprehensive validation ($90.5\% \pm 1.2\%$ mean accuracy)

Key Finding

Models demonstrated remarkable consistency across all splits with less than 2% variation, confirming str

Key Insights from Analysis

Multi-dimensional Fitness

Performance depends on strength, endurance, flexibility, and body composition - not a single factor

Age-Related Decline

Moderate negative correlation between age and performance metrics, particularly grip strength

Body Composition Matters

Lower body fat percentage strongly associated with higher performance classes

Non-linear Relationships

SVM and Neural Networks outperformed linear models, revealing complex patterns in fitness data

Conclusions and Recommendations

Project Achievements

Successfully implemented complete ML pipeline from data exploration to model deployment

Achieved 91% classification accuracy and 85% R^2 for regression

Validated model robustness through comprehensive cross-validation

Identified key predictors of physical performance

Practical Applications

Fitness assessment tools for gyms and health facilities

Early identification of individuals needing targeted training

Personalized training program recommendations

Research platform for fitness and performance studies

Thank You

Questions & Discussion